Classifying quadrilaterals

1

a Rectangle **b** Square **c** Trapezoid **d** Rectangle

e Trapezoid

2

a Rhombus **b** Parallelogram **c** Trapezoid **d** Rhombus

e Trapezoid **f** Parallelogram

3 Parallelogram

4

a False **b** True **c** False **d** False

e False **f** True **g** False **h** True

i False **j** True **k** True **l** False

m False

5

a Rectangle **b** Parallelogram **c** Rhombus **d** Rectangle

e Rhombus **f** Parallelogram **g** Rhombus **h** Parallelogram

6

a Parallelogram **b** Rectangle

7

a There is an additional pair of parallel sides.

b Two pairs of opposite sides are congruent.

c One pair of opposite sides are parallel.

8

a The parallelogram has four congruent sides.

b The parallelogram has four right angles.

c The parallelogram has four congruent sides and four right angles.

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9

a The trapezoid has diagonals that are congruent.

b The trapezoid has base angles that are congruent.

c The trapezoid has opposite angles that are supplementary.

10

To prove: $ABCD$ is a square

	Statements	Reasons
1.	$ABCD$ is a rectangle and a rhombus	Given
2.	$ABCD$ is a parallelogram with four right angles	Definition of a rectangle
3.	$ABCD$ is a parallelogram with four congruent sides	Definition of a rhombus
4.	$ABCD$ is a square	A parallelogram with four congruent sides and four right angles is a square (definition of a square).

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To prove: $ABCD$ is a rectangle

	Statements	Reasons
1.	$ABCD$ is a parallelogram and $\overline{AC} \cong \overline{BD}$	Given
2.	$\overline{AD} \cong \overline{BC}$	If a quadrilateral is a parallelogram, then it has both pairs of opposite sides congruent.
3.	$\overline{AB} \cong \overline{AB}$ and $\overline{CD} \cong \overline{CD}$	Reflexive property of congruence
4.	$\triangle ABD \cong \triangle BAC$ and $\triangle ACD \cong \triangle BDC$	Side-side-side congruence theorem
5.	$\angle ADC \cong \angle BCD$ and $\angle DAB \cong \angle CBA$	Corresponding parts of congruent triangles are congruent (CPCTC).
6.	$\overline{AB} \parallel \overline{DC}$ and $\overline{AD} \parallel \overline{BC}$	Definition of a parallelogram
7.	$\angle ADC$ is supplementary to $\angle BCD$ and $\angle DAB$ is supplementary to $\angle CBA$	If two parallel lines are cut by a transversal, then consecutive interior angles are supplementary (consecutive interior angles theorem).
8.	$m\angle ADC + m\angle BCD = 180^\circ$ and	Definition of supplementary

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	Statements	Reasons
9.	$m\angle ADC + m\angle ADC = 180^\circ$ and $m\angle DAB + m\angle DAB = 180^\circ$	Substitution
10.	$2m\angle ADC = 180^\circ$ and $2m\angle DAB = 180^\circ$	Simplify
11.	$m\angle ADC = 90^\circ$ and $m\angle DAB = 90^\circ$	Division property of equality
12.	$\angle ADC, \angle BCD, \angle DAB$ and $\angle CBA$ are right angles	Definition of a right angle
13.	$ABCD$ is a rectangle	Definition of a rectangle

12

a

To prove: $ABCD$ is a rhombus

	Statements	Reasons
1.	$ABCD$ is a parallelogram and $\overline{AB} \cong \overline{BC}$	Given
2.	$\overline{AB} \cong \overline{CD}$ and $\overline{BC} \cong \overline{DA}$	If a quadrilateral is a parallelogram, then its opposite sides are congruent.
3.	$\overline{AB} \cong \overline{BC} \cong \overline{CD} \cong \overline{DA}$	Transitive property of congruence
4.	$ABCD$ is a rhombus	Definition of a rhombus

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b

To prove: $PQRS$ is a rhombus

	Statements	Reasons
1.	$PQRS$ is a parallelogram and $\overline{PR} \perp \overline{QS}$	Given
2.	$\overline{PR}$ and $\overline{QS}$ bisect each other	If a quadrilateral is a parallelogram, then it has two diagonals that bisect one another.
3.	$\overline{PT} \cong \overline{TR}$ and $\overline{QT} \cong \overline{TS}$	Definition of a bisector
4.	$\angle STP, \angle QTP, \angle QTR, \angle RTS$ are right angles	Definition of perpendicular
5.	$\angle STP \cong \angle QTP \cong \angle QTR \cong \angle RTS$	Definition of congruent angles
6.	$\triangle STP \cong \triangle QTP \cong \triangle QTR \cong \triangle RTS$	Side-angle-side congruence theorem
7.	$\overline{PQ} \cong \overline{QR} \cong \overline{RS} \cong \overline{ST}$	Corresponding parts of congruent triangles are congruent (CPCTC).
8.	$PQRS$ is a rhombus	Definition of a rhombus

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c

To prove: $ABCD$ is a rhombus

	Statements	Reasons
1.	$ABCD$ is a parallelogram and $\overline{BD}$ bisects $\angle ABC$ and $\angle ADC$	Given
2.	$\angle 1 \cong \angle 2$ and $\angle 3 \cong \angle 4$	Definition of an angle bisector
3.	$\overline{AB} \parallel \overline{CD}$ and $\overline{AD} \parallel \overline{CB}$	Definition of a parallelogram
4.	$\angle 1 \cong \angle 3$ and $\angle 2 \cong \angle 4$	If two lines are parallel, then alternate interior angles are congruent (alternate interior angles theorem).
5.	$\angle 1 \cong \angle 4$ and $\angle 2 \cong \angle 3$	Transitive property of congruence
6.	$\overline{DA} \cong \overline{AB}$ and $\overline{BC} \cong \overline{CD}$	If two angles of a triangle are congruent, then the sides opposite those angles are congruent (base angles converse).
7.	$\overline{AB} \cong \overline{DC}$ and $\overline{AD} \cong \overline{BC}$	If a quadrilateral is a parallelogram, then the opposite sides are congruent.
8.	$\overline{AB} \cong \overline{BC} \cong \overline{CD} \cong \overline{DA}$	Transitive property of congruence
9.	$ABCD$ is a rhombus	Definition of a rhombus

13

a

To prove: $ABCD$ is an isosceles trapezoid

	Statements	Reasons
1.	$ABCD$ is a trapezoid, $\angle ADC \cong \angle BCD$, and $\angle APD$ and $\angle BQC$ are right angles	Given
2.	$\angle APD \cong \angle BQC$	All right angles are congruent.
3.	$\overline{AP} \cong \overline{BQ}$	Altitudes of a trapezoid are congruent
4.	$\triangle APD \cong \triangle BQC$	Angle-angle-side congruence theorem
5.	$\overline{AD} \cong \overline{BC}$	Corresponding parts of congruent triangles are congruent (CPCTC)
6.	$ABCD$ is an isosceles trapezoid	Definition of an isosceles trapezoid

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b

To prove: $ABCD$ is an isosceles trapezoid

	Statements	Reasons
1.	$ABCD$ is a trapezoid, $\angle A$ and $\angle C$ are supplementary	Given
2.	$\overline{AB} \parallel \overline{DC}$	Definition of a trapezoid
3.	$\angle A$ and $\angle D$ are supplementary	If two lines are parallel, then consecutive interior angles are supplementary (consecutive interior angles postulate).
4.	$\angle C \cong \angle D$	Supplements of the same angle or congruent angles are congruent (congruent supplements theorem).
5.	$ABCD$ is an isosceles trapezoid	If a trapezoid has base angles congruent, then it is an isosceles trapezoid.

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c

To prove: $ABCD$ is an isosceles trapezoid

	Statements	Reasons
1.	$ABCD$ is a trapezoid, $\overline{AC} \cong \overline{BD}$ and $\overline{AC} \parallel \overline{BP}$	Given
2.	$\overline{AB} \parallel \overline{DC}$	Definition of a trapezoid
3.	$ABPC$ is a parallelogram	Definition of a parallelogram
4.	$\overline{AC} \cong \overline{BP}$	If a quadrilateral is a parallelogram, then its opposite sides are congruent.
5.	$\overline{BD} \cong \overline{BP}$	Transitive property of congruence
6.	$\angle BDC \cong \angle BPC$	If two sides of a triangle are congruent, then the angles opposite those sides are congruent (base angles theorem).
7.	$\angle ACD \cong \angle BPC$	If two lines are parallel, then corresponding angles are congruent (corresponding angles theorem).
8.	$\angle BDC \cong \angle ACD$	Transitive property of congruence
9.	$\overline{DC} \cong \overline{DC}$	Reflexive property of congruence
10.	$\triangle ACD \cong \triangle BDC$	Side-angle-side congruence theorem (SAS)
11.	$\overline{AD} \cong \overline{BC}$	Corresponding parts of congruent triangles are congruent (CPCTC)
12.	$ABCD$ is an isosceles trapezoid	Definition of an isosceles trapezoid

Additional questions

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a Kite

b Rectangle, rhombus, parallelogram

c Rhombus, kite, parallelogram

d Parallelogram

e Kite

f Rectangle, rhombus, kite, parallelogram

g Kite

h Rectangle, rhombus, parallelogram

i Parallelogram

j Trapezoid

k Neither

l Parallelogram

m Trapezoid

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15 3, because otherwise the angle sum would be larger than 360.

16 4, this is the maximum number of angles a quadrilateral can have, and it would give the correct angle sum. Squares and rectangles have four right-angles.

17 Yes, a trapezium.

18 No

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